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Supplementation of isoacids to lactating dairy cows fed low- or high-forage diets: Effects on performance, digestibility, and milk fatty acid profile

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ABSTRACT

Our objective was to determine the effects of isoacids (ISO) on the lactation performance, digestibility, and milk fatty acids (FA) profile of Holstein cows fed 2 forage NDF levels (FL). The study lasted 10 wk (including 2 wk for covariate) using a randomized complete block design. Sixty-four mid-lactating Holstein cows (662 ± 71 kg BW, 119 ± 51 DIM, 2 ± 0.9 parity [±SD]) were blocked by parity, DIM, and prior milk yield (MY) for multiparous cows or genetic merit for primiparous cows, and randomly assigned to 1 of the 4 diets (n = 16). Diets were arranged as a 2 × 2 factorial, with 2 FL containing 21% forage (HF) and 17% forage NDF (LF) without (WIA) or with ISO supplementation (IA; 7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively). Diets were balanced for similar NE_L (1.58 Mcal/kg of DM), CP (16.0%), and total NDF (27.2%). Feed intake and MY were recorded daily. Nutrient digestibility for each cow was determined using indigestible NDF as a marker, and fecal samples were collected at 8-time points (4-h intervals between samples). Individual cow milk samples composited over a 10-wk period were analyzed using GC for FA profile. The statistical model included FL, ISO, and FL × ISO as fixed effects and block as a random effect (*lme4* in R). The ISO did not affect DMI, and LF cows had greater DMI than HF cows (27.8 vs. 26.0 kg/d). However, ISO increased MY (34.7 vs. 37.2 kg/d) and ECM (41.9 vs. 39.0 kg/d) by 7% in cows fed the HF but not in those fed the LF diet, suggesting a FL × ISO interaction. Interestingly, ISO increased ADG (0.4 kg/d) but decreased MUN by 9% only in LF diet as indicated by the FL × ISO interaction. Additionally, ISO increased DM, OM, NDF, and CP digestibility by 10% to 24% in HF, but not in LF (FL × ISO). As expected, ISO increased milk odd-chain FA profiles in the IA groups irrespective of FL; for example, the IA had greater C15:0 (1.87 vs. 1.54 g/100g FA) and a tendency to be greater C17:0 levels (0.86 vs. 0.76 g/100g FA) compared with WIA groups. Overall, ISO improved MY and nutrient digestibility in cows fed the HF diets, whereas it increased ADG and decreased MUN in cows fed the LF diet. Additionally, ISO increased milk odd-chain FA (C15:0 and C17:0) regardless of FL.

Key words: fiber digestibility, milk production, odd-chain fatty acid, branched-chain volatile fatty acids

INTRODUCTION

Forage is an integral component of dairy cattle diets and is utilized by cellulolytic bacteria in the rumen to synthesize volatile fatty acids (**FA**; mostly acetate) and microbial protein. However, despite being a core microbial group in the rumen, cellulolytics thrive on cross-feeding from branched-chain VFA (**BCVFA**) producers, for example, amylolytic bacteria, due to their lack of ability to produce some critical precursors for their growth (Firkins, 2021). Research has shown that the dietary supplementation of specific BCVFA (e.g., isobutyrate, 2-methybutyrate, and isovalerate), either individually or in combinations with another straight-chain FA, valerate (collectively known as isoacids; **ISO**), significantly improves cellulolytic bacterial growth in culture media and in the in vivo rumen environment (Liu et al., 2014; Lee et al., 2021; Roman-Garcia et al., 2021b; Jiang et al., 2023). This acceleration in cellulolytic bacterial growth typically leads to a surge in cellulase activity in rumen (Dai et al., 2015). Moreover, increased cellulase activity facilitates the hydrolysis of glycosidic linkages in cellulose, thereby rendering complex carbohydrates accessible to microbial fermentation (Terry et al., 2019), leading to greater fiber digestion. Consequently, nutrient availability for host animals is enhanced, resulting in

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improved animal performance (e.g., milk production and daily gain; Wang et al., 2019). Additionally, ISO supplementation increased rumen acetate and butyrate levels, which upregulates gene expression required for milk de novo FA synthesis (Liu et al., 2018).

However, the effectiveness of ISO supplementation varies depending on dietary variables such as forage fiber level (**FL**; Firkins, 2021). Typically, a diet with higher FL enhances the relative abundance of primary cellulolytics (e.g., *Fibrobacter succinogenes*, *Ruminococcus albus*, and *Ruminococcus flavefaciens*; Choudhury et al., 2015), and ISO supplementation enhances the rumen BCVFA pool, resulting in improved fiber degradability (Wang et al., 2019). In contrast, low-forage diets compensate for reduced FL with higher nonforage fiber (e.g., soyhulls), which tends to contain more soluble fiber compared with forage fiber and potentially shifts microbial abundance toward more rapidly fermenting bacteria (e.g., amylolytics) at the expense of the slow-growing primary cellulolytics (Choudhury et al., 2015). A recent continuous culture study found that BCVFA supplementation increased total bacterial branched-chain amino acids (**BCAA**) flow (g/d) by 9% in low-forage but 16% in high-forage diets, indicating a competitive BCVFA use in low-forage diets (Mitchell et al., 2023b). Regardless of the disparity in BCVFA utilization depending on FL, increasing the BCVFA pool helps to develop an effective rumen microbial consortium (Moraïs and Mizrahi, 2019).

Therefore, we hypothesized that ISO supplementation in lactating dairy cattle diets would improve fiber digestibility, particularly in high-forage diets compared with low-forage diets. Consequently, this enhanced digestibility could potentially be translated to improved milk production and feed efficiency in a high-forage diet. In contrast, BCVFA supplementation was reported to increase rumen AA and BCAA flow regardless of FL (Mitchell et al., 2023b), leading us to anticipate an increase in milk production in low-forage diets as well. Furthermore, ISO supplementation increases the rumen odd-chain FA (**OCFA**) pool (Roman-Garcia et al., 2021b), which led us to hypothesize that an increased pool might be transmitted to the mammary gland and increase milk OCFA levels (Si et al., 2023; Kupczyński et al., 2024). Therefore, the objectives of this study were to evaluate the response of ISO supplementation on productivity, digestibility, and milk FA profile in mid-lactating dairy cows fed diets with varying levels of forage.

MATERIALS AND METHODS

The study was conducted at the Dairy Research and Training Facility of South Dakota State University (**SDSU**), Brookings, South Dakota. Animal care and the sampling procedure were reviewed by the Institutional Animal Care and Use Committee at SDSU (2202-003A).

Animals, Design of Experiment, and Dietary Treatments

Sixty-four lactating Holstein cows (16 primiparous and 48 multiparous; 662 ± 71 kg BW, 119 ± 51 DIM, 2 ± 0.9 parity; mean ± SD) were used in a 10-wk (2 wk for covariate) randomized complete block design trial. Cows were blocked by parity, DIM, and previous milk production records for multiparous cows or genetic merit for primiparous cows. Cows were randomly assigned to 1 of 4 treatments (n = 16). Diets were arranged as 2 × 2 factorial, with 2 FL containing 21% (**HF**) and 17% forage NDF (**LF**) without (**WIA**) or with ISO supplementation (**IA**, 7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively). The IA was supplied by Zinpro Corporation (Eden Prairie, MN) in the form of calcium salts. In this study, the combination of isobutyrate and 2-methylbutyrate was selected based on recent in vitro findings, which showed that 2-methylbutyrate had the greatest effect on improving NDF digestibility; however, pairing it with either isobutyrate or isovalerate was necessary to maximize NDF digestibility (Roman-Garcia et al., 2021a). Additionally, isobutyrate can be used as a substitute for isovalerate, but isovalerate cannot be used as a substitute for isobutyrate (Mitchell et al., 2023c). The feeding rates of isobutyrate and 2-methylbutyrate in this study were determined by estimating the shortfall of their respective BCAA (valine and isoleucine) supplied by the RDP relative to the estimated bacterial output of those BCAA. Nutritional Dynamic Systems (NDS) Professional V3 software (RUM&N, Reggio Emilia, Italy; https://www.rumen.it/en/ndspro) was used to estimate this shortfall. Details of diet components and composition are available in Table 1 and Table 2. Initially, diets were formulated to meet NASEM (2021) requirements containing similar CP (16.5%), total NDF (28%), and NE_L (1.72 Mcal/kg of DM). However, the composition of the final diets differed marginally from our original formulation, containing, on average 15.8% and 16.2% CP and 26.4% and 28.0% NDF in HF and LF diets, respectively. Similarly, NE_L content was lower but identical in both diets (1.58 Mcal/kg). The individual feed ingredients composition is presented in Supplemental Table S1 (see Notes).

Housing and Management

Cows were housed in a freestall barn with average temperatures ranging from −3.2 to 8.0°C and relative humidity of 55% to 77%, fitted with Calan door systems (American Calan, Northwood, NH) for measuring

Table 1. Ingredients composition of experimental diets

Ingredient, g/100 g of DM	Diet[1]	
	High forage	Low forage
Corn silage	43.9	30.8
Alfalfa hay	12.9	12.7
Alfalfa haylage	8.30	7.96
Cottonseed fuzzy	1.95	3.23
Corn grain	16.4	22.4
Soybean meal, solvent 48% CP	4.78	4.54
AminoPlus[2]	5.00	4.78
Soybean hulls	2.95	10.1
Sodium bicarbonate	1.20	1.10
Limestone, ground	0.59	0.68
Salt, white	0.33	0.32
Urea	0.19	0.19
Calcium phosphate (mono)	0.18	0.17
Magnesium oxide	0.18	0.17
Vitamin mineral premix[3]	0.25	0.23
Megalac[4]	0.96	0.65

[1]High-forage and low-forage diets were fed as control diet without (WIA) or with isoacid supplementation (IA, 7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively). The IA product used was approximately 20% calcium on a weight basis. Based on that value, the additional calcium would be negligible at 8 g or an extra 0.03% of diet DM, approximately.

[2]Rumen ungradable protein source (50% CP with 72% of RUP and 28% RDP; Ag Processing Inc., Omaha, NE).

[3]On the DM basis, the vitamin mineral premix contained 24% calcium, 0.8% phosphorus (inorganic), 0.021% sodium, 0.04% chlorine, 1% potassium, 0.4% magnesium, 0.6% sulfur, 1,600 mg/kg copper, 1,000 mg/kg iron, 80 mg/kg iodine, 4,000 mg/kg manganese, 50 mg/kg selenium, 4,000 mg/kg zinc, 7,056,000 IU/kg vitamin A, 1,764,000 IU/kg vitamin D, 22,050 IU/kg vitamin E.

[4]Rumen bypass fat source (84% fat with 48% C16:0, and 36% C18:1; Volac Wilmar, Hertfordshire, UK).

Table 2. Nutrient composition of experimental diets

Nutrient, g/100 g of DM	Diet[1]	
	High forage	Low forage
DM, %	49.3	55.5
Forage DM, %	65.1	51.5
ADF	18.1	19.7
NDF	26.4	28.0
Forage NDF	20.7	16.6
Nonforage NDF	5.70	11.4
Starch	32.4	30.7
Water-soluble carbohydrates	3.80	3.85
CP	15.8	16.2
RDP	10.3	10.5
RUP	5.53	5.67
Ether extract	4.34	4.37
Total FA	2.74	3.02
Ash	7.95	8.10
Calcium	0.76	0.82
Phosphorus	0.40	0.39
NE_L,[2] Mcal/kg DM	1.58	1.58

[1]High-forage and low-forage diets were fed as control diet without (WIA) or with isoacid supplementation (IA, 7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively).

[2]Calculated value using the equation from NASEM (2021).

individual feed intake. Feed was supplied to individual feedboxes (previous day's intake × 1.1) at 0700 h using a Data Ranger (American Calan). Straw was used as bedding material and was replenished daily at 1700 h. Milking was conducted twice daily at 0500 and 1700 h, using 2 × 8 parallel milking parlors (DeLaval, Des Plaines, IL). Feed was manually moved closer to the cow and replenished if needed before the evening milking. Cows were provided with unrestricted access to clean water throughout the study period.

Sample Collection and Analysis

Individual forage ingredients and TMR were dried at 60°C for 48 h, and grain was dried at 100°C for 24 h for DM determination. Composite ingredient samples were dried at 60°C for 48 h, followed by grinding using a 1-mm sieve in a Wiley mill (Model 3, Arthur H. Thomas Co., Philadelphia, PA) for proximate components and fiber fractions. Proximate components were analyzed according to AOAC International standard protocol: moisture (934.01), ash (942.05), and CP (976.05; AOAC International, 2012). Crude fat was analyzed utilizing American Oil Chemists' Society (**AOCS**) standard procedure (Am 5–04) in a fat extractor (ANKOM XT15 Extractor, Ankom Technology, Macedon, NY; AOCS, 2017). The fiber fractions were analyzed in a fiber analyzer (ANKOM 200 Fiber Analyzer, Ankom Technology) using the AOCS guideline (Ba 6a-05). Therefore, ingredient composition was updated in the NASEM (2021) ration balancing software (NASEM Dairy-8 R2022.09.02; https://drive.google.com/drive/folders/1mbfzZuzfZS_Z8I7I-msB3DfNkrn8eDEX) to determine the TMR composition.

Milk samples were collected into two 50-mL tubes from each cow on a specific day (Tuesday p.m. and Wednesday a.m.). The tube with a preservative (Bronolab-WII; Advanced Instruments Inc., Norwood, MA) was used for milk composition, and the tube without a preservative was used for FA profiling. Milk composition was analyzed weekly using the Milko-Scan FT 6000 (Foss Electric, Hillerød, Denmark). Each week, individual milk FA samples were combined in a 5-mL aliquot, weighted according to respective milk yields (**MY**), and stored at −20°C. Finally, the weekly sample was thawed (4°C) and combined to form a composite sample for each cow, which was analyzed for milk FA profile in a commercial laboratory (Dairyland Laboratories Inc., Arcadia, WI) using GC with flame ionization detection (Baldin et al., 2018).

Individual cows' BW was measured on 2 consecutive days at the start and end of the experimental phase, immediately after the morning milking and before feeding.

The average value of starting and end BW were considered as initial and final BW, respectively. The ADG was calculated from the BW changes, that is, final BW − initial BW, divided by the number of days in the experimental phase. Daily energy expenditure for MY, maintenance, and ADG were calculated using the following NASEM (2021) equations:

$$\text{Milk NE}_L \text{ concentration (Mcal/kg)} =$$
$$9.29 \times \text{kg fat/kg milk} + 5.85 \times \text{kg true protein/kg milk}$$
$$+ 3.95 \times \text{kg lactose/kg milk}$$
$$\text{(NASEM, 2021; equations 3–14b).}$$

This value was multiplied by MY (kg/d) to calculate the daily NE_L utilization for MY (Mcal/d).

$$\text{NE}_L \text{ for maintenance (Mcal/d)} = 0.10 \times \text{BW kg}^{0.75}$$
$$\text{(NASEM, 2021; equations 3–13).}$$

$$\text{NE}_L \text{ for ADG (Mcal/kg ADG)} = 5.6 \text{ Mcal NE}_L\text{/kg ADG}$$
$$\text{(NASEM, 2021; equations 3–19a).}$$

Nutrient digestibility for each cow was determined using indigestible NDF (**iNDF**) as a marker with 8-time points fecal grab samples (4-h interval between samples) collected in wk 9 of the study. In brief, a 50-g fecal grab sample was collected directly from the rectum at each point, stored at −20°C, and composited by animal. The TMR and orts samples were collected on corresponding days and composited by animal. The preparation and chemical analysis of TMR, orts, and fecal samples were conducted similarly to individual forage and TMR analysis described earlier in the "Sample Collection and Analysis" section. Additionally, TMR, orts, and fecal samples were analyzed for 240-h amylase-treated ash-corrected NDF digestibility to determine iNDF content. Feces output was estimated as iNDF intake divided by iNDF concentration in feces (Cochran et al., 1986). Total-tract apparent digestibility was calculated according to Lacey et al. (2020).

Blood samples were collected 4h after feeding from the coccygeal vein during wk 9 of the study using a 10-mL EDTA collection tube (BD Vacutainer, no. 158 USP units, Franklin Lakes, NJ). After collection, the blood sample was immediately centrifuged at 2,000 × *g* for 15 min at 4°C to extract plasma (Megafuge ST1 Plus, Thermo Scientific, Frederick, MD) and stored at −80°C until further analysis. Plasma glucose and insulin were analyzed using glucose (Autokit, Fujifilm Corp., Tokyo, Japan) and bovine insulin ELISA kits (Alpco, Salem, NH), respectively, following the manufacturer's instructions on a BioTek Epoch microplate spectrophotometer (Agilent, Santa Clara, CA).

Statistical Analysis

Data were analyzed using a mixed effects model, including FL, ISO, and FL × ISO as fixed effects and block as a random effect in the *lme4* package in R (version 4.3.1; Bates et al., 2015). Initially, week and parity were considered as fixed effects in the model, but they were removed from the final model because they had no significant effects. Additionally, model assumptions were checked accordingly: quantile-quantile plot for normality of residual, plotting residual against fitted values for checking constant variances, to ensure the maximum fit. All the data were presented as LSM, where $P < 0.05$ was considered statistically significant, and the tendency was declared at $0.05 \leq P < 0.10$. A pairwise group comparison was conducted for any interactions between FL × ISO with $P < 0.10$ using LSM with Tukey's adjustment. Values in tables with different superscript letters differ significantly at $P < 0.05$. Lettering for tended interactions was only applied when at least one intended pairwise comparison (e.g., HF-WIA vs. HF-IA or LF-WIA vs. LF-IA) differed significantly at $P < 0.05$.

RESULTS AND DISCUSSION

Intake, Performance, and Efficiency

Isoacids did not affect DMI ($P = 0.13$), but the LF cows had 7% greater DMI than the HF cows ($P < 0.01$, Table 3). However, ISO exhibited an interaction with FL (FL × ISO) in MY, fat- and protein-corrected milk (**FPCM**), and ECM; that is, ISO supplementation increased production by ~7% in cows fed HF diets ($P < 0.01$) but did not change in those fed LF diets ($P > 0.05$). Furthermore, an additional FL × ISO interaction ($P < 0.01$) was found for ADG, where ISO increased ADG only in cows receiving the LF diet (+0.40 kg/d). However, ISO did not affect feed efficiencies: MY/DMI, FPCM/DMI and ECM/DMI ($P > 0.05$), but the HF cows tended to have greater efficiencies ($P < 0.10$) than the LF cows. In contrast, ISO increased the total net energy used for MY, maintenance, and ADG by 5% (46.8 vs. 44.5 Mcal/d; $P < 0.01$) regardless of FL (Table 3).

Previous research indicated that ISO alters DMI depending on the stage of lactation, that is, no effect was found in mid to late lactation (Liu et al., 2009; Copelin et al., 2021; Mitchell et al., 2023c), whereas DMI was increased in early lactation (Papas et al., 1984; Liu et al., 2018). Because we used mid-lactating dairy cows, no effect on DMI in ISO-supplemented groups is consistent with previous findings. In contrast, ISO was reported to

Table 3. Effect of isoacids on intake and production performance of mid-lactating dairy cows fed varying dietary forage levels

	HF[1]		LF[2]			*P* value[3]		
Variable[4]	WIA[5]	IA[6]	WIA	IA	SEM	FL	ISO	FL × ISO
DMI, kg/d	25.7	26.2	27.5	28.1	0.81	<0.01	0.13	0.87
ADG, kg/d	0.64[c]	0.69[c]	0.93[b]	1.33[a]	0.08	0.03	0.09	<0.01
Yield, kg/d								
MY	34.7[b]	37.2[a]	37.5[a]	38.4[a]	1.68	<0.01	<0.01	0.04
FPCM	35.6[b]	38.2[a]	39.1[a]	38.4[a]	1.14	<0.01	0.06	<0.01
ECM	39.0[b]	41.9[a]	42.8[a]	42.2[a]	1.55	<0.01	0.05	<0.01
Efficiency								
MY/DMI	1.39	1.46	1.39	1.38	0.06	0.09	0.22	0.18
FPCM/DMI	1.44[ac]	1.51[a]	1.46[ab]	1.40[bc]	0.06	0.08	0.78	0.02
ECM/DMI	1.58[ab]	1.66[a]	1.60[ab]	1.53[b]	0.06	0.07	0.79	0.02
Energy expenditure,[7] NE_L MCal/d								
MY (a)	26.7[b]	28.4[a]	29.3[a]	28.8[a]	0.90	<0.01	0.13	0.01
Maintenance (b)	13.2	13.3	13.4	13.5	0.35	0.08	0.14	0.71
ADG (c)	3.88[b]	3.89[b]	5.33[b]	7.71[a]	0.84	<0.01	0.01	0.02
Total (a+b+c)	42.1	44.8	46.9	48.7	1.15	<0.01	0.01	0.59

[a–c]Within a row, mean values with different superscripts differ significantly ($P < 0.05$).

[1]HF = high-forage diet containing 21% forage NDF.

[2]LF = low-forage diet containing 17% forage NDF.

[3]FL = forage level (HF and LF), ISO = isoacids blend supplementation (WIA and IA), and FL × ISO = interaction between FL and ISO.

[4]MY = milk yield; FPCM = fat- and protein-corrected milk, calculated according to IDF (2015), FPCM = milk yield, kg/d × (0.1226 × fat % + 0.0776 × protein % + 0.2534); ECM = energy corrected milk, calculated according to DRMS (2014), ECM = (12.95 × fat yield, kg/d) + (7.65 × true protein yield, kg/d) + (0.327 × milk yield, kg/d).

[5]WIA = no isoacids supplementation.

[6]IA = isoacids supplementation (7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively).

[7]Milk NE_L concentration (Mcal/kg) = 9.29 × kg fat/kg milk + 5.85 × kg true protein/kg milk + 3.95 × kg lactose/kg milk (NASEM, 2021, equations 3–14b); NE_L for maintenance (Mcal/d) = 0.10 × BW $kg^{0.75}$ (NASEM, 2021; equations 3–13); NE_L for ADG (Mcal/kg ADG) = 5.6 Mcal NE_L/kg ADG (NASEM, 2021, equations 3–19a).

increase MY (Wang et al., 2019), but our result indicated an FL × ISO interaction for MY; that is, ISO increased MY, FPCM, and ECM only with the HF diet. This result could be due to ISO increasing NDF and CP digestibility only in HF (Table 4), potentially increasing energy and MP supply for milk production, resulting in improved production (NASEM, 2021).

In contrast, ISO substantially increased ADG in cows fed LF diets as compared with HF diets, indicating altered energy partitioning, diverting more energy toward weight

Table 4. Effect of isoacids on digestibility and plasma metabolites of nutrients in mid-lactating dairy cows fed varying dietary forage levels

	HF[1]		LF[2]			*P* value[3]		
Nutrient[4]	WIA[5]	IA[6]	WIA	IA	SEM	FL	ISO	FL × ISO
Digestibility, %								
DM	67.7[b]	74.8[a]	73.3[a]	74.6[a]	0.92	<0.01	<0.01	<0.01
OM	69.1[b]	75.7[a]	74.2[a]	75.4[a]	0.87	<0.01	<0.01	<0.01
NDF	51.9[b]	64.4[a]	63.4[a]	68.3[a]	2.51	<0.01	<0.01	0.07
CP	68.6[b]	76.5[a]	72.2[a]	74.0[a]	1.46	0.73	<0.01	0.06
Plasma metabolites								
Glu, mg/dL	72.0[ab]	79.0[a]	75.1[ab]	69.9[b]	2.85	0.29	0.76	0.03
Ins, ng/dL	0.51	0.33	0.67	0.53	0.07	0.06	0.09	0.84

[a,b]Within a row, mean values with different superscripts differ significantly ($P < 0.05$).

[1]HF = high-forage diet containing 21% forage NDF.

[2]LF = low-forage diet containing 17% forage NDF.

[3]FL = forage level (HF and LF), ISO = isoacids supplementation (WIA and IA), and FL × ISO = interaction between FL and ISO.

[4]Glu = glucose; Ins = insulin.

[5]WIA = no isoacids blend supplementation.

[6]IA = isoacids supplementation (7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively).

gain rather than milk production (Table 3). Furthermore, we found a 9% reduction in daily enteric methane due to ISO supplementation in LF cows (Redoy et al., 2023), indicating a lower energy loss in the supplemented group, which could potentially be used for weight gain. Previous studies indicated that ISO increased propionate production in cows fed LF diets (Kristensen et al., 2000), which led us to anticipate an upregulated gluconeogenesis—increased glucose production—in the supplemented group in our study. However, we did not observe any change in plasma glucose levels, but insulin levels tended to be reduced as a result of ISO supplementation, suggesting that ISO increased insulin sensitivity in peripheral tissue, resulting in stable plasma glucose concentrations despite upregulated gluconeogenesis (Table 4; Martin and Sauvant, 2007). Consequently, excess propionate in the ISO-supplemented LF group could potentially have been used for lipogenesis in adipose tissue, leading to the accumulation of fat and an increase in ADG (Martin and Sauvant, 2007). Furthermore, past research has shown that maintaining a positive energy balance (e.g., gaining weight) during late lactation leads to improved performance in the subsequent lactation (Mishra et al., 2016). Given that ISO supplementation increased ADG in cows fed the LF diet, it could potentially boost the energy supply of mid-to-late lactation cows, which in turn could exert a positive effect on subsequent lactations. Nevertheless, except for numerical differences, feed efficiencies (e.g., MY/DMI, FPCM/DMI, and ECM/DMI) were comparable ($P > 0.05$) between the WIA and IA groups in the HF diet, even though ISO increased yield by ~7% without affecting DMI.

Nutrient Digestibility and Plasma Metabolites

Isoacids increased DM, OM, NDF, and CP digestibility by 10% to 24% depending on the variable in the HF diet ($P < 0.01$; Table 4). Isoacid supplementation did not affect these variables in the LF diet (i.e., FL × ISO; $P > 0.05$). As expected, the LF diet had greater DM, OM, and NDF digestibility than HF ($P < 0.01$), and CP digestibility was comparable between FL ($P > 0.05$). Furthermore, ISO did not affect plasma glucose levels ($P = 0.76$) but tended to reduce insulin levels ($P = 0.09$) regardless of FL. Additionally, plasma insulin level tended to be higher in LF than HF ($P = 0.06$), but glucose level was comparable between FL ($P = 0.29$).

Increased OM and NDF digestibility in the HF diet due to ISO supplementation was consistent with previous studies (Liu et al., 2009; Jiang et al., 2023). Although our study observed increased CP digestibility in the ISO-supplemented group in HF, Jiang et al. (2023) reported no change, and Kamble and Thakur (2004) noted an 8% increase in CP digestibility. This discrepancy likely results from dietary variations: Jiang et al. (2023) used a diet with 50% forage, compared with 80% in Kamble and Thakur (2004). Furthermore, ISO could not improve the DM, OM, CP, and NDF digestibility in the LF diet, which was in line with previous findings (Copelin et al., 2021). Increased NDF digestibility in the ISO-supplemented group in the HF diet might be due to an increase in the rumen cellulolytic bacterial population and their enzyme activity in supplemented group (Liu et al., 2014; Wang et al., 2015). Moreover, a recent study confirmed that NDF digestibility was positively associated with increasing cellulolytic taxa in the rumen, which further validates the higher NDF digestibility in the ISO-supplemented group in the HF diet (Jiang et al., 2023). Furthermore, these cellulolytic microbiomes utilize rumen ammonia as a source of nitrogen for their growth, which could be a reason for improving CP digestibility in the ISO-supplemented group in the HF diet (Mitchell et al., 2023b). Increased DM and OM digestibility might be driven predominantly by the increase in NDF digestibility in the HF ISO-supplemented group. In LF cows, the rumen microbial ecosystem shifted toward amylolytic bacteria over cellulolytic bacteria, reducing the overall requirement for ISO (Firkins and Mitchell, 2023). This could explain why NDF digestibility remained unchanged after adding ISO to the LF diet in our study, aligning with the findings of others (Wang et al., 2019; Mitchell et al., 2023c).

Isoacids did not change plasma glucose levels but tended to reduce insulin levels in the supplemented group (main effect, Table 4). Prior research highlighted that ISO increases plasma glucose levels, contrary to our findings. The results of this study do align with previous findings that indicated ISO supplementation reduced insulin levels (Liu et al., 2009; Wang et al., 2017). In the HF diet, ISO was speculated to increase rumen acetate production without affecting propionate production (Jiang et al., 2023), suggesting that both WIA and IA groups might have a similar propionate pool for gluconeogenesis, leading to the same blood glucose levels. As mentioned earlier, ISO might improve energy balance in LF cows, translating into greater ADG. This might increase insulin sensitivity, resulting in less insulin being required for glucose uptake (De Koster et al., 2015).

Milk Composition and Components Yield

Milk composition was not affected by FL and ISO, except that ISO supplementation decreased preformed FA ($P = 0.02$) irrespective of FL (main effect), and decreased mixed FA ($P < 0.01$) and MUN ($P = 0.01$) in the LF diet (i.e., FL × ISO interaction; Table 5). Furthermore, ISO increased fat, total FA, protein, and total solids yield by 7% to 9% in HF cows ($P < 0.01$) and reduced fat and total FA yield by 5% to 10% in LF cows (FL × ISO; P

Table 5. Effect of isoacids on milk composition and component yield of mid-lactating dairy cows fed varying dietary forage levels

	HF[1]		LF[2]			*P* value[3]		
Variable[4]	WIA[5]	IA[6]	WIA	IA	SEM	FL	ISO	FL × ISO
Milk composition, %								
True protein	3.39	3.46	3.45	3.46	0.07	0.25	0.15	0.33
Fat	4.21	4.24	4.30	4.03	0.17	0.49	0.14	0.08
Total FA	3.97	3.99	4.05	3.80	0.16	0.51	0.14	0.09
De novo FA[7]	1.09	1.11	1.12	1.06	0.05	0.81	0.36	0.08
Preformed FA[7]	1.37	1.32	1.39	1.29	0.06	0.73	0.02	0.41
Mixed FA[7]	1.55[a]	1.58[a]	1.57[a]	1.47[b]	0.06	0.21	0.32	0.04
SNF	9.21	9.23	9.20	9.20	0.10	0.69	0.86	0.86
Lactose	4.70	4.65	4.65	4.64	0.05	0.24	0.31	0.33
MUN (mg/dL)	12.4[a]	12.4[a]	13.0[a]	11.8[b]	0.49	0.89	0.01	0.01
Component yield, kg/d								
True protein	1.17[b]	1.27[a]	1.28[a]	1.31[a]	0.06	<0.01	<0.01	0.03
Fat	1.44[c]	1.55[b]	1.69[a]	1.52[b]	0.07	0.05	0.07	<0.01
Total FA	1.35[c]	1.45[b]	1.50[a]	1.42[b]	0.06	0.04	0.75	<0.01
SNF	3.20	3.41	3.43	3.51	0.15	<0.01	<0.01	0.10
Lactose	1.64	1.72	1.74	1.79	0.08	<0.01	<0.01	0.31
TS	4.63[b]	4.95[a]	5.03[a]	5.02[a]	0.19	<0.01	0.01	<0.01

[a–d]Within a row, mean values with different superscripts differ significantly ($P < 0.05$).

[1]HF = high-forage diet containing 21% forage NDF.

[2]LF = low-forage diet containing 17% forage NDF.

[3]FL = forage level (HF and LF), ISO = isoacids blend supplementation (WIA and IA), and FL × ISO = interaction between FL and ISO.

[4]FA = fatty acid; SNF = solids not fat.

[5]WIA = no isoacids supplementation.

[6]IA = isoacids supplementation (7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively).

[7]Near-infrared spectroscopy-based analysis.

< 0.01). Milk components yield was 4% to 7% greater ($P < 0.05$) in LF diets than in HF diets due to greater MY in LF cows than HF cows (Table 3).

Previous studies found that ISO had no effect on milk composition except for milk fat and true protein. However, this shift in milk fat and protein percentage was inconsistent; for instance, ISO has been reported to have no effect (Mitchell et al., 2023c), somewhat decrease (Liu et al., 2009), tend to increase (Liu et al., 2018), or significantly increase (Copelin et al., 2021) milk fat content, depending upon the study. Different dietary scenarios might be a factor in variable effects on milk fat (%) content because the rumen fermentation profile is highly diet dependent. A diet with higher FL produces more acetate in the rumen, which could be utilized for milk fat synthesis in the mammary gland (Walker et al., 2004). Furthermore, increasing NDF digestibility is speculated to increase acetate production in the rumen (Jiang et al., 2023). Although, in our study, ISO increased NDF digestibility in the HF diet, this did not translate into increased milk fat content (Table 5). The probable explanation would be that ISO increased MY by 7% rather than fat content (i.e., dilution effect), which resulted in a +0.11 kg/d increase in total milk fat yield in the HF diet supplemented with ISO. In the case of LF, the milk fat percentage did not change, but the total fat yield was reduced by 0.17 kg/d (Table 5). As mentioned earlier, ISO might have shifted microbial populations or altered microbial metabolic pathways, which ultimately reduced total fat yield in the supplemented group. Moreover, a recent study using an in vitro continuous culture system reported that ISO decreased total bacterial AA-N by 2% of total N in HF but increased it by 4% in LF (Mitchell et al., 2023b), which might explain why ISO supplementation decreased MUN in LF. However, the same author did not observe a drop in MUN when animals were fed a diet with ~53% forage DM (Mitchell et al., 2023c). In contrast, Liu et al. (2020) found the ISO supplementation reduced rumen ammonia N by 5% in a diet containing 50% forage on a DM basis, which could partially explain the MUN reduction in our study for the ISO-supplemented group in the LF diet.

Milk Fatty Acid Profile

Isoacids supplementation increased milk C4:0 in HF but decreased it in LF diets (FL × ISO; $P = 0.04$), without affecting other short-chain FA (**SCFA**; C4:0–C10:0; $P > 0.05$). As expected, ISO increased OCFA profiles in the supplemented group irrespective of FL; the supple-

Table 6. Effect of isoacids on individual milk fatty acid profile of mid-lactating dairy cows fed varying dietary forage levels

	HF[1]		LF[2]			*P* value[3]		
Fatty acid, g/100 g	WIA[4]	IA[5]	WIA	IA	SEM	FL	ISO	FL × ISO
C4:0	0.01[c]	0.08[b]	0.19[a]	0.00[d]	0.10	0.45	0.37	0.04
C6:0	2.76	2.68	2.74	2.76	0.06	0.92	0.21	0.75
C8:0	1.62	1.60	1.63	1.63	0.04	0.25	0.51	0.95
C10:0	3.97	4.05	4.09	4.16	0.13	0.25	0.51	0.95
C12:0	4.61	4.76	4.76	4.96	0.18	0.19	0.21	0.84
C14:0	12.9	13.1	12.7	13.0	0.27	0.41	0.27	0.74
C14:1 *cis*-9	1.34	1.33	1.23	1.41	0.08	0.69	0.27	0.17
C15:0	1.48	1.65	1.59	1.90	0.14	0.09	0.03	0.53
C16:0	34.9	34.9	34.2	34.6	0.73	0.37	0.64	0.72
C16:1 *trans*-9	0.34	0.31	0.34	0.34	0.02	0.41	0.53	0.24
C16:1 *cis*-9	2.14	2.04	2.03	2.20	0.11	0.76	0.71	0.09
C17:0	0.70	0.84	0.81	0.87	0.04	0.49	0.05	0.60
C18:0	8.23	8.04	8.53	7.98	0.52	0.77	0.39	0.65
C18:1 *trans*-6	0.27	0.27	0.27	0.27	0.01	0.77	0.91	0.88
C18:1 *trans*-9	0.22	0.22	0.21	0.22	0.01	0.48	0.39	0.15
C18:1 *trans*-10	0.42	0.48	0.42	0.51	0.04	0.68	0.04	0.81
C18:1 *trans*-11	0.62	0.62	0.63	0.53	0.44	0.19	0.11	0.14
C18:1 *cis*-6	0.38	0.27	0.26	0.32	0.06	0.39	0.61	0.06
C18:1 n-9	18.6	17.9	17.5	17.2	0.52	0.03	0.19	0.62
C18:1 n-7	0.69	0.73	0.69	0.72	0.04	0.91	0.15	0.87
C18:2 *trans*-9, *trans*-12	0.09	0.08	0.12	0.12	0.04	0.09	0.78	0.62
C18:2 n-6	2.15	2.49	3.32	2.74	0.61	0.13	0.82	0.32
C19:0	0.17	0.17	0.17	0.18	0.01	0.99	0.29	0.32
C20:0	0.12	0.08	0.30	0.07	0.12	0.35	0.17	0.29
C20:1 *trans*-11	0.12	0.12	0.12	0.11	0.01	0.38	0.16	0.67
C20:1 n-11	0.44	0.39	0.50	0.51	0.03	<0.01	0.48	0.27
C20:3 n-6	0.14	0.13	0.14	0.14	0.01	0.28	0.73	0.96
C22:1	0.20	0.21	0.22	0.22	0.01	0.02	0.42	0.92
Σ CLA	0.37	0.46	0.38	0.49	0.10	0.79	0.24	0.94
Σ Trans FA	1.74	1.76	1.76	1.74	0.06	0.95	0.98	0.77

[a–d]Within a row, mean values with different superscripts differ significantly ($P < 0.05$).

[1]HF = high-forage diet containing 21% forage NDF.

[2]LF = low-forage diet containing 17% forage NDF.

[3]FL = forage level (HF and LF), ISO = isoacids blend supplementation (WIA and IA), and FL × ISO = interaction between FL and ISO.

[4]WIA = no isoacids supplementation.

[5]IA = isoacids supplementation (7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively).

mented group had greater C15:0 (1.87 vs. 1.54 g/100 g of FA; $P = 0.03$) and tended to have greater C17:0 levels (0.86 vs. 0.76 g/100 g of FA; $P = 0.05$). Although ISO slightly increased the C18:1 *trans*-10 FA concentration in the supplemented group from 0.42 to 0.50 g/100 g of FA, supplementation had no effect on the total *trans* fatty acid level (Σ *trans* FA; Table 6; $P = 0.98$). Nevertheless, milk FA profiles did not change much with FL, except for C18:1 n-9, which was greater in HF ($P = 0.03$), and C20:1 n-11, and C22:1, which were greater in LF ($P < 0.05$).

Prior studies found that ISO increased the acetate: propionate ratio, whereas their effect on rumen butyrate was inconsistent, ranging from no change to a slight increase (Liu et al., 2009, 2020; Lee et al., 2021). Furthermore, butyrate directly contributes to milk C4:0, whereas other SCFA are synthesized from acetyl-CoA, potentially influenced by rumen acetate production or β-oxidation of FA (Glasser et al., 2008). We hypothesized that ISO might change all SCFA, mostly in HF diets, but our study only showed an increase in C4:0. Our results align with those of Lee et al. (2021) but contradict those of Liu et al. (2018), who found an increase in all SCFA (C4:0–C10:0) with ISO supplementation. One potential explanation for the increase in C15:0 and C17:0 in the milk of the ISO-supplemented group could be because of the catabolism of ISO, which increases the metabolic pool of propionyl-CoA that can then be used by the mammary gland to synthesize these FA (Abdoul-Aziz et al., 2021). Though isobutyrate supplementation was reported to increase propionyl-CoA, implying increased propionate production (Kristensen et al., 2000); supplementation with a blend of BCVFA (e.g., ISO) either linearly decreased (Liu et al., 2009) or had no effect (Lee et al., 2021) on rumen propionate concentrations.

Another potential explanation for the ISO supplementation increasing C15:0 and C17:0 is that the ISO are absorbed directly from the rumen and then converted to their respective CoA in the liver and used by the mammary gland for FA formation (Liu et al., 2018, 2020). Data are limited to ISO metabolism in the bovine liver and hepatic conversion of ISO to the respective CoA.

A third possible explanation for the increase in C15:0 and C17:0 with ISO supplementation is that rumen microbes use ISO for the formation of OCFA (e.g., C15:0 and C17:0) to maintain their cellular integrity and functionality (Lee et al., 2021). Eventually, the fat and protein components of these microbes would be available for cows to utilize, including for milk fat. Previous research reported that ISO increased cellulolytic bacterial activity by facilitating OCFA formation, which improved cell fluidity (Wu and Palmquist, 1991). However, it is unlikely that they directly supply C15:0 and C17:0 to the intestine. Recent research could not detect any difference in bacterial C15:0 and C17:0 composition as a result of ISO supplementation (Lee et al., 2021; Mitchell et al., 2023a). It is speculated that OCFA from rumen microbes are degraded enzymatically in the cow's intestine or further modified in the liver (e.g., β-oxidation) or in the mammary gland (e.g., C15:0 and C17:0 biosynthesis).

Several studies have demonstrated that C15:0 and C17:0 FA have the potential to improve human health, for example, by reducing the risk of Type 2 diabetes (Pfeuffer and Jaudszus, 2016; Risérus and Marklund, 2017; Prada et al., 2021) and cardiometabolic diseases (Venn-Watson and Schork, 2023). In addition, research has shown that C15:0, an active and exogenous FA, has mitochondrial-reparative, anti-inflammatory, and antifibrotic characteristics, which leads to reduced inflammation, anemia, and liver fibrosis along with lowering plasma cholesterol, triglycerides, and glucose levels (Venn-Watson and Schork, 2023). Because ISO supplementation increased milk C15:0 and tended to increase C17:0, it could potentially be added to dairy diet to enhance milk quality regardless of FL.

CONCLUSIONS

In the HF diet, ISO supplementation increased MY, FPCM, and ECM by ~7% and nutrient digestibility by 10% to 24%, whereas in the LF diet, ISO increased ADG by 0.40 kg/d. However, ISO did not affect the DMI or feed efficiency in either diet. Furthermore, ISO decreased MUN by 10% in the LF diet only, suggesting a potential reduction in urinary N excretion because MUN and urinary N excretion are strongly correlated, which warrants additional research focusing on nitrogen footprint. Moreover, ISO increased milk OCFA profiles (e.g., C15:0 and C17:0) in the supplemented group, regardless of FL. It would be worth investigating the effects of ISO supplementation on the ruminal gene expressions using meta-transcriptomics or proteomics profile under varying FL. In addition, the response to ISO supplementation in divergently low or highly efficient cows on enteric CH_4 emissions and ruminal gene expression warrants further investigation.

NOTES

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Nonstandard abbreviations used: AOCS = American Oil Chemists' Society; BCAA = branched-chain amino acids; BCVFA = branched-chain VFA; FA = fatty acids; FL = forage NDF level; FPCM = fat- and protein-corrected milk; GC = gas chromatography; HF = high-forage diet containing 21% forage NDF; IA = diet with isoacid supplementation, 7.85 mmol/kg of DM and 3.44 mmol/kg of DM for isobutyrate and 2-methylbutyrate, respectively; iNDF = indigestible NDF; ISO = isoacids, collective term for BCVFA with straight-chain valerate; LF = low-forage diet containing 17% forage NDF; MY = milk yield; OCFA = odd-chain FA; SCFA = short-chain FA; SDSU = South Dakota State University; WIA = diet without isoacid supplementation.

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